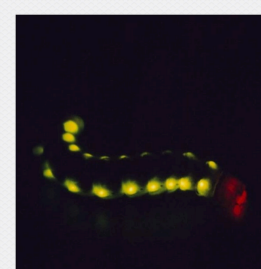
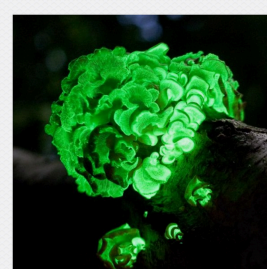
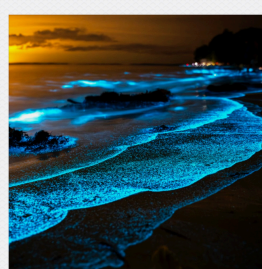
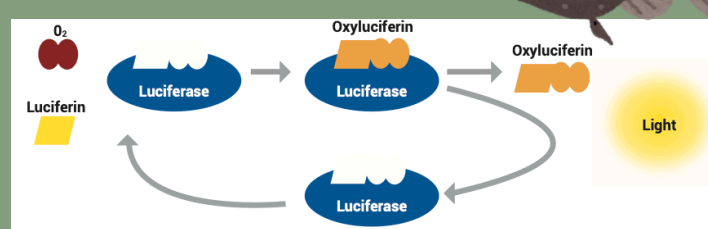


Bioluminescence imaging (BLI)

By: Kanthida Philavong, Leow Jing Qi, Tong Yi

WHAT IS BIOLUMINESCENCE?

- molecule called luciferin(light emitting compound, substrate) react with oxygen to produce light.
- luciferase(enzyme) catalyzes reaction between luciferin and oxygen.
- reaction releases energy in form of light with minimal heat.



WHAT IS BIOLUMINESCENCE IMAGING (BLI)?

It is a technique that harnesses bioluminescence and sensitive cameras to track biological processes and biomolecules in real time. However, it is still under preclinical research.

HOW DOES BLI WORK?

Bioluminescence imaging is an in-vivo characterization and quantification of molecular and cellular processes. BLI turns chemical energy into light or photons by mixing enzymes such as luciferases or photoproteins with luciferin substrates, increasing the visibility of single tumor cells.

MAIN CHALLENGE:

Bioluminescence signals are faint and get absorbed/scattered by tissues.

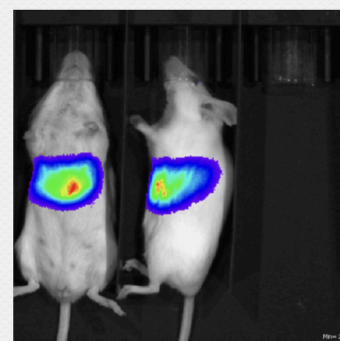
SOLUTIONS:

- developing brighter luciferin.
- develop more advanced ways to gather more light.
- high light sensitivity cameras.
- Using red luciferin for deeper tissue penetration.

PROS AND CONS OF BLI

Pros:

- high sensitivity.
- non-invasive.
- real time monitoring.
- cellular-level insight.
- improved diagnosis.



Cons:

- Limited depth penetration.
- Requires genetic modification.
- signal variability.
- long exposure timings.



POTENTIAL APPLICATIONS

- Preclinical research.
- Medication development.
- Tumor research and tracking.
- examining biological processes.
- understanding cancer metabolism.

CONCLUSION

Although it has not been fully implemented in clinical settings, bioluminescence imaging is a promising technique that provides great potential benefits to the future of healthcare. It is a low-cost, simple-to-implement method for studying biological processes, including cancer, infection, and diabetes.

SEE OUR REFERENCES HERE:

